ECSE 318 Lab 4, Problem 2: Shift-Add Multiplier

Group 9. Structural 4-bit VHDL model, compiled and simulated in ModelSim Intel FPGA Starter Edition 2020.1.

Compilation: vcom -2008 adder.vhdl dff.vhdl MultiplierVHDL.vhdl MultiplierVHDL_tb.vhdl

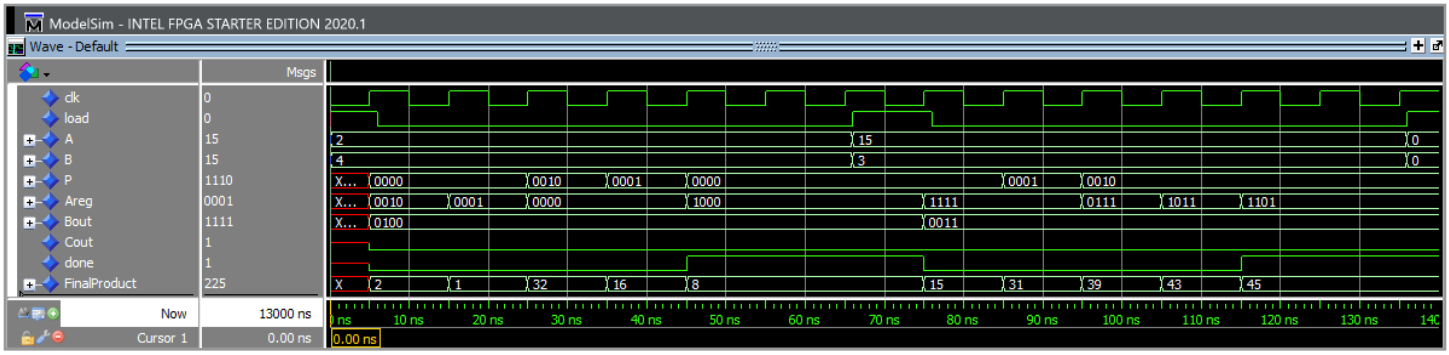
```
Model Technology ModelSim - Intel FPGA Edition vcom 2020.1 Compiler 2020.02 Feb 28 2020
Start time: 17:34:55 on Nov 01,2026
-- Compiling entity adder
-- Compiling architecture Behavioral of adder
-- Compiling entity dff
-- Compiling architecture Behavioral of dff
-- Compiling entity MultiplierVHDL
-- Compiling architecture Behavioral of MultiplierVHDL
-- Loading entity adder
-- Loading entity dff
-- Compiling entity MultiplierVHDL_tb
-- Compiling architecture sim of MultiplierVHDL_tb
-- Loading entity MultiplierVHDL
End time: 17:34:55 on Nov 01,2026, Elapsed time: 0:00:00
Errors: 0, Warnings: 0
```

Simulation transcript

```
# vsim -c -do "run -all; quit -f" MultiplierVHDL_tb
# Start time: 17:34:57 on Nov 01,2026
# Loading work.multiplierVHDL_tb(sim)
# Loading work.multiplierVHDL(behavioral)
# Loading work.adder(behavioral)
# Loading work.dff(behavioral)
# run -all
# ** Note: 2 * 4 = 8
#    Time: 46 ns  Iteration: 0  Instance: /multiplierVHDL_tb
# ** Note: 15 * 3 = 45
#    Time: 116 ns  Iteration: 0  Instance: /multiplierVHDL_tb
# ** Note: failures: 0
#    Time: 12936 ns  Iteration: 0  Instance: /multiplierVHDL_tb
# End time: 17:34:57 on Nov 01,2026, Elapsed time: 0:00:00
# Errors: 0, Warnings: 0
```

The testbench runs the two required cases, then all 256 4-bit input pairs. "failures: 0" means every product was correct and took exactly 4 clocks.

Waveform, 0 to 140 ns (empty rows and the side panel are cropped out of the ModelSim window)



Case	FinalProduct after each clock	Result
2 * 4	2 (loaded), 1, 32, 16, 8	8, done goes high at 45 ns
15 * 3	15 (loaded), 31, 39, 43, 45	45, done goes high at 115 ns